Set up NAT Network

1️⃣ Create NAT Network

In VirtualBox, under Network --> NAT Network --> Create

Name: e.g. `3BHITS` (you can decide, name doesnt matter)

IPv4 Prefix: `10.0.5.0/24`

Go to `Port Forwarding`

- Add a rule
  - Host Port: `4448` (if you already have that port in another rule, choose another, e.g. `4449`)
  - Guest IP: `10.0.5.10`
  - Guest Port: `22`

2️⃣ Select NAT Network in VM

Under VM settings, go to Network an change `attached to` to

```
NAT Network
```

Boot VM again

Set up DHCP Server

Important Information: Log into `root` user on the VM (`su -`)

1️⃣ Update system

```
apt update && apt upgrade -y
```

2️⃣ Change interface to static

```
vi /etc/network/interfaces
```

Change this line (or comment it out):

```
iface enp0s3 inet dhcp
```

To this:

```
iface enp0s3 inet static
        address 10.0.5.--/24
        gateway 10.0.5.1
```

Restart networking.service:

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```
systemctl restart networking.service
```

Check if ip is static:

```
ip a
```

3️⃣ Install Kea DHCP Server

```
apt install kea-dhcp4-server -y
```

4️⃣ Edit DHCP config

> replace the .conf file with the one provided

5️⃣ Enable DHCP server

```
systemctl enable kea-dhcp4-server # not nessecary
```

```
systemctl restart kea-dhcp4-server
```

6️⃣ Check the status of the DHCP server

```
systemctl status kea-dhcp4-server
```

> needs to be `active (running)`

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